Geometric Matrix of Kochenov: A Synthesis

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Содержание

The AU–Day–$\sqrt{3}$ Light Relation: Two Geometric Observations

Abstract

This note presents two independent numerical relations connecting the astronomical unit (AU), the mean solar day (86 400 s), the speed of light c, and the geometric factor $\sqrt{3}$.

Constants

$$\begin{aligned}
AU &= 149\,597\,870\,700 \text{ m (exact IAU definition)} \\
c &= 299\,792\,458 \text{ m/s (defined SI value)} \\
\text{Day} &= 86\,400 \text{ s} \\
\sqrt{3} &= 1.73205080756887729352744634150587236694280525381038
\end{aligned}$$

First Relation: Geometric Approximation of c

$$c_{\text{cube}} = \frac{100 \cdot AU \cdot \sqrt{3}}{86400}$$

Numerical evaluation yields:

$$c_{\text{cube}} \approx 299\,897\,121.246 \text{ m/s}$$

Comparison with the defined value:

$$\Delta c = c_{\text{cube}} - c \approx 104\,663.246 \text{ m/s}, \qquad \frac{\Delta c}{c} \approx 3.491 \times 10^{-4} \ (0.03491\%)$$

Thus, the combination $\frac{100 \cdot AU \cdot \sqrt{3}}{86400}$ approximates the speed of light to within 0.035%.

Second Relation: Near-Identity

Define:

$$X = \frac{100 \cdot AU}{86400} \approx 173\,145\,683.680556 \text{ m/s}, \qquad Y = \frac{c}{\sqrt{3}} \approx 173\,085\,256.327320 \text{ m/s}$$

The absolute difference is:

$$\Delta = X - Y \approx 60\,427.353 \text{ m/s}$$

The relative deviation is:

$$\frac{\Delta}{Y} \approx 3.491 \times 10^{-4} \ (0.03491\%)$$

Light-Travel Time Interpretation

$$t_{AU} = \frac{AU}{c} \approx 499.004784 \text{ s}, \qquad \frac{86400}{100\sqrt{3}} \approx 498.830633 \text{ s}, \qquad \Delta t \approx 0.17415 \text{ s}$$

Conclusion

Two clear numerical observations emerge:

1. $\frac{100 \cdot AU \cdot \sqrt{3}}{86400}$ approximates c with 0.0349% accuracy.
2. $\frac{100 \cdot AU}{86400}$ and $\frac{c}{\sqrt{3}}$ differ by 60 427 m/s, same relative deviation.

The Cancellation of the Metric Scale in $GM_\odot \approx 31^6 r_{AU}$

A geometric parametrisation of the solar gravitational parameter is based on the discrete anchor 31 and the astronomical unit r_{AU}. The Sun–Earth–Moon radius ratio defines the orbital speed scale. The conventional factor 10^3 associated with km/s cancels when the expression is written consistently in SI units. In normalised form, $GM_\odot \approx 31^6 r_{AU}$, where 31 is a geometric seed. The deviation from the CODATA value is about 0.04%.

Basic Relation

$GM_\odot = v_\oplus^2 \times r_{AU}$.

Earth's Orbital Speed from Geometry

$$\Gamma = \frac{R_\odot \times R_{\mathrm{Moon}}}{R_\oplus^2} \approx 29.7788, \qquad \Gamma \approx \frac{31^3}{1000}.$$

In SI units: $v_\oplus = \Gamma$ km/s $\approx \frac{31^3}{1000} \times 10^3$ m/s $= 31^3$ m/s. Hence $v_\oplus^2 \approx 31^6$.

Solar GM

$GM_\odot = v_\oplus^2 \times r_{AU} \approx 31^6 r_{AU}$.

Numerical Value

$31^6 = 887\,503\,681$, $r_{AU} = 1.495978707 \times 10^{11}$ m, $GM_\odot \approx 1.3275 \times 10^{20}$ m^3/s^2. CODATA: 1.3271244×10^{20}, deviation $\sim 0.04\%$.

Conclusion

In this parametrisation, the metric factor cancels, leaving the dimensionless geometric structure unchanged. The integer 31 acts as a discrete anchor extracted from the Earth–Moon–Sun radius ratio and the golden-ratio spiral seed $S = \phi^6 \times 31$.

The Nine-Planet Solar System: A Kochenov Geometric Framework without Newtonian Gravitational Constant

Abstract

A single geometric seed $S = \phi^6 \cdot 31$ generates the geometric speed of light, the base length of the Solar System, orbital radii and speeds of nine planets, Earth's orbital speed, and the solar gravitational parameter $GM_\odot$ without fitted constants.

The Geometric Seed S

$$\phi = \frac{1+\sqrt{5}}{2}, \qquad S = \phi^6 \cdot 31 \approx 556.272429 \text{ s}$$

Geometric Speed of Light

$$c_{\text{geom}} = S^3 \cdot \sqrt{3} \cdot \left(1 + \frac{4}{720}\right) \approx 2.99798395 \times 10^8 \text{ m/s}$$

CODATA value: 2.99792458×10^8 m/s, deviation +0.002%.

Base Length L_{base}

$$L_{\text{base}} = c_{\text{geom}} \cdot S \approx 1.66766773 \times 10^{11} \text{ m} \approx 1.114766 \text{ AU}$$

Origin of Γ and Earth's Speed

$$\Gamma = \frac{R_\odot \cdot R_{\text{Moon}}}{R_\oplus^2} \approx 29.7788, \qquad v_\oplus = \Gamma \cdot 1 \text{ km/s} \approx 29.78 \text{ km/s}$$

Observed: 29.7847 km/s (deviation 0.02%).

Kochenov Bridge: 31^3 to Γ

$$\frac{31^3}{1000} \approx \Gamma, \qquad 31^3 = 29791, \qquad 1000 \cdot \Gamma \approx 29779 \text{ (deviation 0.02\%)}$$

Planetary Operators

$$\begin{array}{rlrlrl}
\text{Mercury}: & \dfrac{\sqrt{3}}{5} L_{\text{base}}, & \text{Venus}: & \dfrac{360}{S} L_{\text{base}}, & \text{Earth}: & 1.000, \\
\text{Mars}: & \dfrac{\sqrt{3}}{\sqrt[3]{2}} L_{\text{base}}, & \text{Jupiter}: & \dfrac{8\sqrt{3}}{3} L_{\text{base}}, & \text{Saturn}: & 5\sqrt{3} L_{\text{base}}, \\
\text{Uranus}: & 10\sqrt{3} L_{\text{base}}, & \text{Neptune}: & \dfrac{31\sqrt{3}}{2} L_{\text{base}}, & \text{Pluto}: & \dfrac{31\sqrt{3}}{\sqrt{2}} L_{\text{base}}.
\end{array}$$

Complete Nine-Planet Architecture

Planet	Theor AU	Real AU	Theor km/s	Real km/s
Mercury	0.386	0.387	47.9	47.9
Venus	0.721	0.723	35.0	35.0
Earth	1.000	1.000	29.78	29.78
Mars	1.532	1.524	24.12	24.07
Jupiter	5.149	5.203	13.06	13.07
Saturn	9.654	9.537	9.64	9.69
Uranus	19.309	19.190	6.80	6.80
Neptune	29.928	30.110	5.43	5.43
Pluto	42.323	39.482	4.58	4.67

Conclusion: The Nine-Planet Manifest

The number 720 provides a synchronization lock:

$$86400 = 120 \times 720, \quad c = S^3\sqrt{3}\left(1+\frac{4}{720}\right), \quad \frac{\phi^{12}}{\sqrt{5}} = 144 + \frac{1}{720}, \quad \frac{m_n}{m_p} = 1 + \frac{1}{720}.$$

Speed of Light (m/s)

Core (integers only)

$$c_{\text{core}} = 144^3 \times 100 = 298\,598\,400 \text{ m/s}, \qquad c_{\text{SI}} = 299\,792\,458 \text{ m/s}, \qquad \Delta = 0.398\%$$

Koch (geometric)

$$c_{\text{geom}} = (\phi^6 \cdot 31)^3 \cdot \sqrt{3} \cdot \left(1 + \frac{4}{720}\right) \approx 2.99798395 \times 10^8 \text{ m/s}$$

Deviation from c_{SI}: $\approx 0.002\%$.

The Omega Node and Integer Core Approximation of the Speed of Light

Аннотация

Within the Kochenov geometric framework, the speed-of-light scale emerges through two independent numerical constructions built from the same set of discrete operators. One path originates from the small vacuum residue Δ, generating a resonance node Ω. The second arises from a purely integer projection of the central Fibonacci anchor 144. Both constructions approach the physical speed of light with nearly identical accuracy ($\sim 0.4\%$), despite following entirely different internal pathways. No derivation of c is claimed; these relations are recorded as structural correspondences within the framework.

1. The Foundational Residue

The framework is grounded in the identity

$$\frac{\phi^{12}}{\sqrt{5}} = 144 + \frac{1}{720} + \Delta, \qquad \Delta \approx -1.33956 \times 10^{-8}.$$

Δ represents the geometric tension between continuous golden-ratio geometry and the discrete integer lattice.

2. The Omega Node

A dimensionless resonance node is defined as

$$\Omega = \frac{1}{\sqrt{31(-\Delta)}} \approx 1552.6.$$

3. Quadratic Projection

Quadratic amplification scaled by $5^3 = 125$ produces

$$c_\Omega \approx 125 \cdot \Omega^2 \approx 301\,013\,477 \text{ m/s}.$$

Relative deviation from $c = 299\,792\,458$ m/s: $\sim 0.4\%$.

4. Integer Core Approximation

Independently of Δ, a purely discrete projection based on the Fibonacci anchor gives

$$c_{\text{core}} = 144^3 \times 100 = 298\,598\,400 \text{ m/s},$$

with a relative deviation of 0.398%.

5. Structural Observation

Two distinct internal routes converge toward the same macroscopic velocity scale:

- **Route A (Resonance Projection):** $\Delta \to \Omega \to 125 \cdot \Omega^2 \to c_\Omega$
- **Route B (Integer Projection):** $144 \to 144^3 \times 100 \to c_{\text{core}}$

Despite originating from different parts of the matrix (one involving the vacuum tension Δ, the other using only integer operators), both yield values within $\sim 0.4\%$ of the observed speed of light.

6. Conclusion

The Kochenov framework contains two independent numerical structures that naturally generate values close to the physical speed of light. The first relies on the vacuum residue and resonance amplification, while the second emerges directly from the integer lattice. The recurrence of the speed-of-light scale from the core operators (31, 144, 5, Δ) is a noteworthy structural pattern. Whether this reflects a deeper geometric mechanism or an internal symmetry of the construction remains an open question, but these consistent numerical correspondences deserve archival preservation within the broader geometric matrix.

The Gravitational Node in the Kochenov Matrix

1. Topological Anchor and Elimination of the Continuous Representation

From the Binet representation of the 12th Fibonacci square $F_{12} = 144$, we introduce the numerical identity:

$$\frac{\phi^{12}}{\sqrt{5}} = 144 + \frac{1}{720} + \Delta, \qquad \Delta \approx -1.3396 \times 10^{-8}.$$

Within the proposed framework, this relation serves as a bridge between a continuous representation (based on the golden ratio ϕ) and a discrete lattice built from the integers 144 and 720. When substituted into the expression for the gravitational constant, the irrational blocks ϕ^{12} and $\sqrt{5}$ cancel, leaving only the discrete parameters and the residual term Δ.

2. Rational Form of the Gravitational Operator

The starting expression in SI units is:

$$G_{\text{Koch}} = \frac{31^2\sqrt{5}}{\phi^{12} \cdot 10^{11}} \text{ m}^3\text{kg}^{-1}\text{s}^{-2}.$$

Using the identity above, the irrational factors cancel, and we obtain:

$$G_{\text{Koch}} = \frac{31^2}{144 + \frac{1}{720} + \Delta} \times 10^{-11}.$$

The zeroth-order approximation ($\Delta = 0$) gives the arithmetic core:

$$G_{\text{core}} = \frac{31^2}{144} \times 10^{-11} = \frac{961}{144} \times 10^{-11} = 6.6736111\ldots \times 10^{-11}.$$

Including the residual term Δ yields the refined value:

$$G_{\text{Koch}} = 6.67354674496161812375919977179042360791 32 \times 10^{-11} \text{ m}^3\text{kg}^{-1}\text{s}^{-2}$$

The full precision value is:

$$\begin{aligned} G_{\text{Koch}} = {} & 6.673546744961618123759199771790423607913 2 \\ & 17454871515565576576721833255940989010244791334067576927 3 \times 10^{-11} \end{aligned}$$

3. Comparison with Experimental Measurements

For reference, one of the most precise recent laboratory determinations reported by the NIST group of Stephan Schlamminger (April 2026) is:

$$6.67387 \pm 0.00038 \times 10^{-11}.$$

The difference between the refined geometric value and the experimental result is approximately 0.0048%. The geometric value lies within the experimental uncertainty range.

4. Embedding into the Einstein Coupling Term

The Einstein coupling factor $8\pi G$ becomes:

$$8\pi G_{\text{Koch}} = 8\pi \cdot \left(\frac{31^2}{144 + \frac{1}{720} + \Delta} \right) \times 10^{-11}.$$

In the zeroth-order approximation ($\Delta = 0$):

$$8\pi G_{\text{core}} = 8\pi \cdot \left(\frac{961}{144} \right) \times 10^{-11}.$$

5. Geometric G: From Earth's Geometry to G

From Earth-based definitions: 1 $\mathrm{m}^3/1$ kg $\approx 10^3$ (to 0.003%). Applying this to G_{Koch}:

$$G_{\text{Earth}} = G_{\text{Koch}} \times 10^3 = \frac{31^2\sqrt{5}}{\phi^{12}} \times 10^{-8}\ \mathrm{s}^{-2}.$$

6. Comparison with Laboratory Measurements

Source	$G \times 10^{-11}$	Gap from G_{Earth}
Schlamminger (2014)	6.6729 ± 0.0005	+0.010%
G_{Earth}	$6.67354674496\ldots$	—
CODATA 2018	6.67430 ± 0.00015	−0.011%

The Kochenov Length and Mass: A Geometric Invariant Pair

A dimensionless geometric length scale and its reciprocal mass scale are constructed from the discrete invariants of the Kochenov matrix: 31, 144, 720, and the residual $\Delta \approx -1.33956 \times 10^{-8}$. Unlike the conventional Planck length, which is tied to the SI system, the Kochenov length is a pure geometric invariant, independent of anthropic units.

1. From the Gravitational Node to Length and Mass

The geometric gravitational node is:

$$G_{\text{geom}} = \frac{31^2}{144 + \frac{1}{720} + \Delta}.$$

By analogy with the Planck construction (where $\ell_P = \sqrt{G}$ in natural units), we define:

$$L_{\text{Koch}} = \sqrt{G_{\text{geom}}} = \frac{31}{\sqrt{144 + \frac{1}{720} + \Delta}}.$$

The corresponding mass scale is its reciprocal:

$$M_{\text{Koch}} = \frac{1}{L_{\text{Koch}}} = \frac{\sqrt{144 + \frac{1}{720} + \Delta}}{31}.$$

These two quantities satisfy $L_{\text{Koch}} \cdot M_{\text{Koch}} = 1$, reflecting the natural duality between length and mass in units where $\hbar = c = 1$.

2. Numerical Value

Ignoring the residual Δ:

$$L_{\text{core}} = \frac{31}{\sqrt{144 + \dfrac{1}{720}}} \approx \frac{31}{12.0000578} \approx 2.58332088.$$

The dominant term is simply $\frac{31}{12} = 2.583333\ldots$, with the $1/720$ providing a tiny correction. Including Δ leaves the value virtually unchanged (≈ 2.583325).

3. Golden-Ratio Representation

Using the identity $\dfrac{\phi^{12}}{\sqrt{5}} = 144 + \dfrac{1}{720} + \Delta$, the Kochenov length becomes:

$$L_{\text{Koch}} = \frac{31}{\sqrt{\phi^{12}/\sqrt{5}}} = \frac{31 \cdot \sqrt[4]{5}}{\phi^6}.$$

This form expresses the same invariant solely through the golden ratio ϕ, $\sqrt{5}$, and the integer 31.

4. Interpretation

- L_{Koch} is a geometric invariant of the matrix, not just a number.
- M_{Koch} is its reciprocal, reflecting the natural duality of length and mass in natural units.
- Unlike the Planck length (1.616255×10^{-35} m), which is tied to the SI system, L_{Koch} contains no anthropic scales.

Thus, the Kochenov framework provides a pure, self-contained geometric pair that replaces the conventional Planck scale, free from human unit conventions.

$$\boxed{L_{\text{Koch}} = \frac{31}{\sqrt{144 + \frac{1}{720} + \Delta}}, \qquad M_{\text{Koch}} = \frac{1}{L_{\text{Koch}}}}$$

Kochenov Time: The Geometric Planck Time

In natural units where $c = 1$ and $\hbar = 1$, time and length are dimensionally equivalent. Therefore, once the geometric Planck length is defined, the Planck time follows directly:

$$t_P^{\text{geom}} = \ell_P^{\text{geom}} = \frac{31}{\sqrt{144 + \dfrac{1}{720} + \Delta}}, \qquad \Delta \approx -1.33956 \times 10^{-8}.$$

Numerical value:

$$t_P^{\text{geom}} \approx \mathbf{2.583325}$$

(in natural geometric units of the Kochenov Matrix).

Interpretation

The Planck time is no longer an exotic consequence of three independent constants ($\hbar$, G, c). It emerges as a pure geometric ratio — the same discrete step of the matrix that defines the fundamental length.

Just as the Kochenov Length is the minimal spatial cell of the lattice, the Kochenov Time is the minimal temporal tick of that same lattice. Both are derived from the same fundamental anchors: the integer 31, the Fibonacci node 144, and the synchronization constant 720, with the tiny residual Δ representing the "breathing"of the system.

Relation to the Conventional Planck Time

In the conventional SI system:

$$t_P = \sqrt{\frac{\hbar G}{c^5}} \approx 5.391 \times 10^{-44}\ \text{s}.$$

In the Kochenov framework, this value is not fundamental. The fundamental quantity is the geometric ratio $t_P^{\text{geom}} \approx 2.583325$. The SI value emerges only when this ratio is projected through the anthropic scales $\hbar$, G, and c.

$$\boxed{t_{\text{Koch}} = \frac{31}{\sqrt{144 + \frac{1}{720} + \Delta}}}$$

The Einstein–Hilbert Action in the Planck Limit

In general relativity the Einstein–Hilbert action is given by

$$S_{EH} = \frac{c^3}{16\pi G} \int R\sqrt{-g}\, d^4x,$$

where R is the Ricci scalar and g is the determinant of the metric tensor.

When working in natural Planck units ($\hbar = 1$, $c = 1$, $G = 1$), the action simplifies to the standard dimensionless form

$$S_{EH} = \frac{1}{16\pi} \int R\sqrt{-g}\, d^4x.$$

Within the Kochenov Geometric Framework the gravitational constant is replaced by its geometric inverse expressed through the discrete invariants of the matrix:

$$G_{\text{geom}} = \frac{31^2}{144 + \frac{1}{720} + \Delta}, \qquad \Delta \approx -1.3396 \times 10^{-8}.$$

Substituting the inverse of this geometric coupling into the Einstein–Hilbert prefactor yields

$$S_{EH} = \frac{1}{16\pi} \left(\frac{144 + \frac{1}{720} + \Delta}{31^2} \right) \int R\sqrt{-g}\, d^4x.$$

This expression does not alter the mathematical structure of the Einstein–Hilbert action. Instead, it provides an explicit parameterization of the gravitational coupling constant in terms of the fundamental discrete numbers of the Kochenov matrix (31, 144, 720) together with the small residual tension Δ.

The resulting form connects the smooth geometry of general relativity with the discrete lattice underlying the Kochenov framework, offering a bridge between continuous spacetime curvature and a deeper integer-based structure.

The same discrete parameters (31, 144, 720, Δ) appear in independent parametrizations of the speed of light, the fine-structure constant, the neutron–proton mass ratio, and the cosmological constant within the same framework.

Whether this correspondence reflects a fundamental geometric origin of gravity or remains a useful numerical reformulation is left as an open question for further investigation.

Microworld

mn/mp = 1 + 1/720

Relative difference: 0.00105%

$$\frac{m_n}{m_p} \approx 1 + \frac{1}{720}$$

Numerically: $1 + 1/720 = 1.001388888\ldots$
CODATA: $1.00137841946\ldots$
Relative difference: 0.00105%

The Electron as a Transitional Scaling Node

The neutron–proton mass difference and the electron mass satisfy a simple numerical relation:

$$\frac{m_n - m_p}{m_e} \approx 2.530985, \qquad \frac{5}{2} + \frac{31}{1000} = 2.531,$$

with a relative deviation of about 0.0006%. The same mass difference is numerically close to the lattice step:

$$m_n - m_p \approx 0.001388449 \text{ u}, \qquad \frac{1}{720} \approx 0.001388889.$$

Thus, the electron acts as a calibration node bridging the baryonic mass asymmetry (governed by 720) and the discrete anchor 31, converting the nucleonic splitting into a stable mass scale that defines atomic structure.

Baryon Excess and Electron Mass: A Numerical Relation

$$\frac{m_n - m_p}{m_e} \approx 2.53098, \qquad \frac{5}{2} + \frac{31}{1000} = 2.531, \qquad \text{deviation } 2 \times 10^{-5}.$$

Conclusion

The ratio of the baryon mass excess to the electron mass is numerically close to the simple expression $5/2 + 31/1000$. The integers 5, 31, and 1000 appear in other constructions.

Electron Scaling Relation

A compact approximation for the electron–proton mass ratio is:

$$m_e \sim m_p \times \frac{1}{720} \times \frac{1}{2.531} \times (1 - \alpha),$$

where

$$2.531 = \frac{5}{2} + \frac{31}{1000}, \qquad \alpha^{-1} \approx 137.$$

Thus, the electron mass emerges from the proton mass modulated by three structural operators:

- the synchronization node 720,
- the electron scaling node 2.531,
- the electromagnetic correction $(1 - \alpha)$.

Numerically:

$$\frac{m_p}{720 \cdot 2.531} \approx 0.5148, \qquad (1 - \alpha) \approx 0.9927007,$$

$$m_e \approx 0.51099 \text{ MeV},$$

which coincides with the experimental value to within a relative deviation of about 10^{-6}.

Appendix A: The Three Integer Relations

Let $\phi = \frac{1+\sqrt{5}}{2}$ be the golden ratio. The core of the framework consists of three expressions built from the integers 31 and 144 and the decimal scale 100/1000:

$$31^3 \approx 1000 \cdot v_\oplus \qquad \text{(Orbital Node)} \tag{1}$$

$$\frac{31^2}{144} \approx G \times 10^{11} \qquad \text{(Gravitational Node)} \tag{2}$$

$$144^3 \times 100 \approx c \qquad \text{(Electromagnetic Node)} \tag{3}$$

The Synchronization Lock 720

The three core relations are tied together by the highly composite number $720 = 6!$:

- Earth's rotation: $86400\text{ s} = 120 \times 720$
- Speed of light correction: $c_{\text{geom}} = S^3 \times \sqrt{3} \times (1 + 4/720)$, where $S = \phi^6 \times 31$
- Golden identity: $\phi^{12}/\sqrt{5} = 144 + 1/720$
- Neutron-proton mass ratio: $m_n/m_p = 1 + 1/720$

Thus, the same integer 720 appears in terrestrial time, electromagnetism, pure geometry, and particle physics, indicating a unified discrete lattice.

Appendix B: A High-Precision Cosmological Identity Linking $\phi^{12}/\sqrt{5}$, Fibonacci Node F_{12}, and the Structural Synchronization Lock 720

$$\frac{\phi^{12}}{\sqrt{5}} = 144 + \frac{1}{720} + \Delta, \qquad \Delta = -1.3396 \times 10^{-8}, \qquad \frac{|\Delta|}{X} \approx 9.30 \times 10^{-11}.$$

Appendix C: Recurring Operators of the Kochenov Geometric Matrix

The framework repeatedly employs a small set of discrete operators that appear across multiple constructions. These operators serve as structural building blocks rather than independently fitted parameters.

Operator	**Structural Role**
31	Spatial anchor (Mersenne prime $2^5 - 1$)
144	Fibonacci lattice node (F_{12}, the only square in Fibonacci sequence)
720	Synchronization operator (6!, spinor cycle)
Δ	Vacuum tension residue ($\approx -1.3396 \times 10^{-8}$)
ϕ	Continuous golden-ratio geometry
5	Golden radical source ($\sqrt{5}$)

Derived Quantities

The same operators reappear in the construction of several physical scales.

Structural Observation

A notable feature of the framework is that the same small collection of operators — 31, 144, 720, Δ, ϕ, and 5 — reappears in constructions associated with gravitational, cosmological, atomic, and temporal scales.

The framework does not claim that these correspondences constitute established physical derivations. Rather, it records a recurring numerical structure whose persistence across multiple scales motivates further investigation.

Quantity	Geometric Expression
Gravitational coupling (G_{Koch})	$\dfrac{31^2}{144 + \frac{1}{720} + \Delta} \times 10^{-11}$
Speed of light (c_{cube})	$\dfrac{100 \cdot AU \cdot \sqrt{3}}{86400}$
Kochenov length (L_{Koch})	$\dfrac{31}{\sqrt{144 + \frac{1}{720} + \Delta}}$
Kochenov time (t_{Koch})	L_{Koch} (natural units)
Omega node (Ω)	$\dfrac{1}{\sqrt{31(-\Delta)}}$
Neutron-proton mass ratio	$1 + \dfrac{1}{720} + \Delta \times 10^3$
Electron scaling	$m_e \sim m_p \times \dfrac{1}{720} \times \dfrac{1}{2.531} \times (1 - \alpha)$

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